CS-UY 1134 — Homework 4

Questions 1 and 2

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Question 1

Recursive ideas

sum_lst1: The base case returns the only element in a one-element list. Each recursive call sums the slice that starts at index 1, then adds the first element.

sum_lst2: The base case returns lst[low] when low = high. Each recursive call advances low, then adds lst[low].

1. sum_lst1 recursion tree

```
Problem size      Local cost
    n               Θ(n)   (creates lst[1:])
    |
   n-1              Θ(n-1)
    |
   n-2              Θ(n-2)
    ...
    1               Θ(1)
```

$T(n) = T(n - 1) + \Theta(n) = \Theta(n + (n-1) + ... + 1) = \Theta(\mathbf{n^2})$.

2. sum_lst2 recursion tree

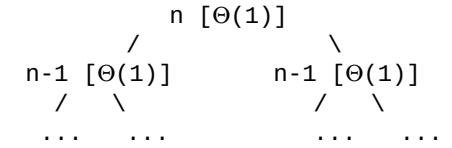
```
Range size        Local cost
    n               Θ(1)
    |
   n-1              Θ(1)
    |
   n-2              Θ(1)
    ...
    1               Θ(1)
```

$T(n) = T(n - 1) + \Theta(1) = \Theta(\mathbf{n})$.

Conclusion: sum_lst2 is asymptotically faster: $\Theta(n)$, compared with $\Theta(n^2)$ for sum_lst1.

Question 2

(a) fun1




```
Level i: 2^i calls, each with local cost Θ(1)
Depth: n; leaves: 2^n
```

T(n) = 2T(n – 1) + Θ(1) = Θ**(2^n)**. The cost of level i is Θ(2^i), not constant.

(b) fun2

```
Input             Local cost
 n                 Θ(1)
 |
 n/2               Θ(1)
 |
 n/4               Θ(1)
 ...
 1                 Θ(1)
 |
 0                 Θ(1)
```

There is one call per level and Θ(log n) levels. Thus T(n) = T(n/2) + Θ(1) = Θ**(log n)**.

(c) fun3

```
Input             Local cost
 n                 Θ(n)
 |
 n/2               Θ(n/2)
 |
 n/4               Θ(n/4)
 ...
 1                 Θ(1)
 |
 0                 Θ(1)
```

The costs form the geometric series n + n/2 + n/4 + ... + 1 < 2n. Thus T(n) = T(n/2) + Θ(n) = Θ**(n)**.